An Unidentifiability Oracle for H&E → Spatial Transcriptomics

ML Lab Research Proposal — trustworthy selective prediction for virtual spatial transcriptomics

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1. Problem Statement

Predicting spatial gene expression from routine H&E histology promises cheap, slide-wide molecular maps, but current models give point predictions with no trustworthy notion of when to believe them. A weak prediction–truth correlation is ambiguous: the gene may be *biologically unpredictable* from morphology, or the measurement may simply be *noisy / dropout-corrupted* — and existing uncertainty and abstention methods conflate the two. We build an **unidentifiability oracle** that separates intrinsic biological unpredictability from technical measurement noise, so a selective-prediction layer abstains on genuinely unidentifiable genes rather than merely noisy ones.

2. Proposed Approach

We **train a high-capacity H&E→ST predictor** (TRIPLEX / BLEEP / STFlow-class) end-to-end, fine-tuning an **ungated** pathology backbone (CTransPath / DINOv2) on the WSI tiles. Per-gene unidentifiability U is the deep-ensemble out-of-fold aleatoric residual **minus** an externally measured noise floor (a true Xenium technical replicate) **minus** registration-induced variance; a learned **dual-head** oracle (prediction + variance head, "HYBRID") is then calibrated by post-hoc spatial-Mondrian conformal for coverage-controlled selective risk. **Higher predictor capacity is the point:** a stronger, GPU-trained f leaves only genuinely irreducible residual, tightening U toward true unidentifiability — a frozen, under-powered predictor over-attributes its own under-fitting to "biology." We chose this over Bayesian UQ (uncalibrated) and naive variance abstention (tracks dropout, not biology) because the aleatoric/epistemic split is not recoverable from a predictor's own outputs; it needs the external clean reference and replicate Xenium uniquely provides. The contribution is the oracle — the predictor class is wrapped, not claimed.

3. Dataset

All data are public. The primary source is GEO **GSE243280** (Janesick 2023): paired breast **Xenium** (two serial technical replicates, GSM7780153/4), **Visium**, and post-stain **H&E**, on a 313-gene panel. We extend to five co-registrable organ triads (breast, colon, lung, liver/HCC, ovary), including a Xenium-5K panel for a scaling test — on the order of hundreds of thousands of cells per section. Preprocessing: niche-level spatial binning, serial-section rigid registration into a shared frame, panel-gene intersection, and WSI tiling for backbone fine-tuning.

4. Evaluation Metrics

- **Selective-risk–coverage curve** (selective risk vs. retained fraction) — headline.
- **Separation gap:** AUROC(deferral↔high-U) − AUROC(deferral↔dropout), ≥ 0.10, holding under two independent predictor architectures.
- **Marginal coverage** within each Mondrian (gene×region) group, within ±2% of nominal $1-\alpha$.
- **Noise-floor clearing** (genes with reproducible signal above the replicate floor); **retained-fraction × utility** ≥ 15%; **cross-panel scaling gap** (v1 → Xenium-5K); trained-f vs. frozen-baseline tightening of U.

5. Compute Requirements

GPU-intensive (cloud H100 via Modal). Two stages: (1) **fine-tune the ungated backbone + train the dual-head predictor** on WSI tiles, with a **deep ensemble across architectures** for the epistemic term — on the order of tens of GPU-hours across five organs, plus the throughput-bound one-time tiling + embedding-extraction pass (Xenium-5K is the largest); (2) lightweight post-hoc conformal on CPU. Mixed precision, gradient checkpointing, and embedding caches keep each backbone within a single H100; ensembling and the organ panel scale out elastically on Modal. Fits the 192 GB budget. Dependencies: PyTorch, **Modal**, `timm`, scanpy/anndata, MAPIE/crepes.

6. Expected Deliverables

- **Trained HYBRID dual-head oracle** + a fine-tuned **ungated** backbone (no gated-model dependency), with the frozen post-hoc oracle as a reproducible fallback baseline.
- A reusable **benchmark** for biological-vs-technical separation on H&E→ST, released with the code.
- Selective-risk–coverage results across the organ panel + the panel-scaling ablation; Modal training pipeline.
- A short paper targeting **WACV 2026**, pre-registered and fully reproducible (per-run git SHA + config hash).

7. Risk & Mitigation

- **Signal-vs-noise floor.** Addressed by a **solved pre-flight**: a cheap, no-training control on the real breast data already shows the technical-replicate noise floor clears at niche resolution, with biologically coherent markers (ESR1, keratins, SFRP1) leading. The staged design gates all downstream work behind this control.
- **Training stability of the dual head.** Staged with a **guaranteed fallback** — the frozen post-hoc oracle is a complete, always-available result, and the trained HYBRID is an upgrade layered on top, so a shippable deliverable always exists. Variance-head training uses β-NLL and gradient surgery for stability.
- **Serial-section registration.** Handled at niche (not single-cell) resolution, using 10x-provided H&E↔Xenium homographies plus a built-in perturbation-sensitivity check; residual error stays in the conservative, signal-shrinking direction.
- **Reproducibility & engineering.** Version-controlled, test-gated, and already passed an independent multi-model code review — engineering risk to the timeline is low.